

Data Integrity Report

Problem 1: Duplicates

- What I found : Duplicates
- Where it was : All columns (Symmetrical Duplicates)
- How big it was: 8
- What I did and why: Drop the symmetric duplicates cause they are 100% a duplicated information not a new data

Problem 2 : Nulls

- What I found: Nulls
- Where it was: -first_name , last_name , grade , neighborhood , membership_status , join_date , checkout_id , return_date , publication_year , book_id , checkout_date , genre , pages , publisher

- How big it was:
 - Total(351)
 - - first_name (5)
 - - last_name (5)
 - - grade (43)
 - - neighborhood (5)
 - - membership_status (5)
 - - join_date (12)
 - - checkout_id (44)
 - - book_id(18)
 - - checkout_date(18)
 - - genre(18)
 - - pages(18)
 - - return_date (109)
 - - publication_year (33)
 - - publisher (18)

- What I did and why:

I filled (or partially filled) the following columns :

-grade: Using normalize to fill data proportionally so don't lead to a huge increase
In specific record or Intentional bias like if I used mod

-neighborhood: I filled it as unknown while I could use mode but it's not real and we
It could lead data to be biased and it could be fabricating to the data

-join_date: I filled most of it, not the whole of it . As some data are ghost (orphan). I filled using That typically the student won't checkout unless he has joined so i used the date of join

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-checkout_id: I used the highest value in check_id to increment starting from it and I used a special mask to avoid filling 18 members that I mentioned in the data I didn't fill .

-publication_year : I used the median of publication_year of the publisher and if there isn't any other books for this publisher

I use the overall median of publishing of the dataset. I used both Ways ensure that I can get an accurate approximate

-return_date : I used the information that they will not return these books till summer end Which means by the end of the summer which is 31 of august

- **Data that I didn't fill or partially didn't fill:**

- first_name 05
- last_name 05
- join_date 06
- checkout_id 18
- book_id 18
- checkout_date 18
- return_date 65
- Borrowing_count 0
- genre 18
- pages 18
- publication_year 18
- publisher 18

I have 18 member with no checkouts and incase I filled these data I'm making checkouts out Out of nowhere for . They didn't borrow anything so I can't just fill them .

Join_date could be just guessed . using mode will lead to a guess not an approximate number to the real one and same thing for return_date for this data.

First name and last name kept without filling or deleting for reasons mentioned in Problem 4

Problem 3 : Data Inconsistency

- What I found: Data Inconsistency
- Where it was: neighborhood , membership_status
- How big it was: 5 (Maddi , zamalek , Naser city , Naser city , Heliopolis) , 2(active , inactive)

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- What I did and why: I found that some data like neighborhood and membership_status had inconsistency.

Like Heliopolis and HELIOPOLIS , active and Active So i used str.lower and title and strip to check that there's no spaces or difference due to a capital or small letter and title for neighborhoods . Also Naser city with double spaces which I had to use replace to resolve it

Problem 4 : Unmatched IDs

- What I found: Unmatched IDs
- Where it was: member id , first_name
- How big it was: 3 (1104,1201,1150)
- What I did and why: These were students from the Summer Reading Kickoff HTML file who didn't exist in the main library database. I kept their checkout records so we don't lose borrowing data, but I had to leave their demographic information (like first_name) as null because we cannot invent or fabricate names for IDs that don't exist in the system.